

Lecture #11, Oct 21, 2025

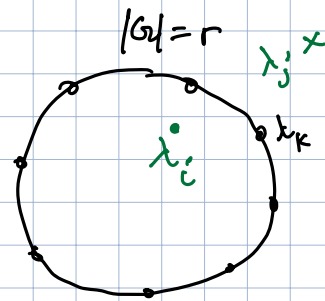
spectral methods for solving PDEs

- Goal: i) spectral discretization: linear & nonlinear cases
ii) Identify the system's spectrum from numerical solution

Today: spectral discretization for linear case.

Remark.

$$G = |G| e^{i\theta}$$
$$\text{or } z = r e^{i\theta} \Rightarrow$$



For any FD scheme.

lect # 10 $\Rightarrow$

$$u^{n+1} = A u^n$$
$$u^n = A u^{n-1}$$

A is amplification matrix equivalent to amplification factor G.

spectral discretization

linearly independent set:

A set $\{\phi_1, \phi_2, \dots, \phi_N\}$ is linearly independent

$$\text{if } \sum_{n=1}^N c_n \phi_n = 0 \Rightarrow c_1 = c_2 = \dots = c_N = 0$$

Basis: For a vector space $\mathcal{U}$

$$\text{if for any } u \in \mathcal{U}, \text{ we have } u(x) = \sum_{n=1}^N c_n \phi_n(x)$$

We say $\{\phi_n\}_{n=1}^N$ is a basis for V .

Remark

1) $\Phi = \begin{bmatrix} \vdots & \vdots & \dots & \vdots \\ \phi_1 & \phi_2 & \dots & \phi_N \\ \vdots & \vdots & \dots & \vdots \end{bmatrix}$ is a matrix.

Independent $\Rightarrow \text{rank}(\Phi) = \text{number of columns}$

Basis $\Rightarrow \det(\Phi) \neq 0$

spectral method for linear PDE

Represent $u(x, t)$ as

$$u(x, t) = \sum_{n=-\infty}^{+\infty} \hat{u}_n(t) \phi_n(x)$$

Here $\{\phi_n(x)\}_{n=-\infty}^{\infty}$ is infinite dimensional basis

$\{\hat{u}_n(t)\}$ contains corresponding weights or coefficients

Numerical method requires a truncated sum

$$u_N = \sum_{n=-N/2+1}^{N/2} \hat{u}_n(t) \phi_n(x)$$

$$\left| \begin{array}{l} N = 4 \\ -N/2 = -2 \\ -2, -1, 0, 1, 2 = N/2 \\ \Rightarrow 5 \end{array} \right.$$

Want $u_N(x, t)$ to be the best approximation of $u(x, t)$

So, we minimize L^2 error $\|u(x, t) - u_N(x, t)\|_2$

This gives

$$\hat{u}_n(t) = \frac{\langle u(x, t), \phi_n(x) \rangle}{\|\phi_n(x)\|^2}, \quad \text{for } n = -N/2+1, \dots, N/2$$

Where $\langle u(x, t), \phi_n(x) \rangle = \int u(x, t) \phi_n(x) dx$ is inner product.

Orthogonality

We say $u(x)$ and $v(x)$ orthogonal

$$\text{if } \langle u, v \rangle = \int_a^b u(x) v(x) dx = 0$$

For any set $\{\phi_n\}$, we need

$$\langle \phi_i, \phi_j \rangle = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}$$

$\{\phi_n\}$ is orthonormal set

$$\text{if } \langle \phi_i, \phi_j \rangle = \delta_{ij}$$

$$\text{or } \int_a^b \phi_i(x) \phi_j(x) dx = \delta_{ij}$$

Observation

$$u(x, t) = \dots + \hat{u}_0(t) \phi_0(x) + \hat{u}_1(t) \phi_1(x) + \dots$$

$$\text{then } \langle u(x, t), \phi_0 \rangle = \langle \cdot, \phi_0 \rangle + \langle \cdot, \phi_0 \rangle + \dots + \langle \hat{u}_0 \phi_0, \phi_0 \rangle + \langle \hat{u}_1 \phi_1, \phi_0 \rangle + \dots$$

$$\begin{aligned} \langle \hat{u}_0 \phi_0, \phi_0 \rangle &= \int \hat{u}_0(t) \phi_0(x) \phi_0(x) dx \\ &= \hat{u}_0(t) \underbrace{\int \phi_0(x) \phi_0(x) dx}_{\| \phi_0 \|^2} \\ &= \hat{u}_0(t) \| \phi_0 \|^2 \end{aligned}$$

$\| \phi_0 \| = 1$ for orthonormal basis

$$\langle \hat{u}_0 \phi_0, \phi_1 \rangle = \hat{u}_0 \int \phi_0(x) \phi_1(x) dx = 0$$

$$\text{Hence } \langle \hat{u}_n(t) \phi_n(x), \phi_n(x) \rangle = \hat{u}_n(t) \| \phi_n(x) \|^2$$

$$\Rightarrow \langle u(x,t), \phi_n(x) \rangle = \hat{u}_n(t) \|\phi_n(x)\|^2$$

$$\boxed{\hat{u}_n(t) = \frac{\langle u(x,t), \phi_n(x) \rangle}{\|\phi_n\|^2}}$$

Q: Derive the best approximation coefficients $\hat{u}_n(t)$ for a given basis.

Fourier spectral method

$$\text{let } \phi_n(x) = e^{2\pi i n x / L} \quad \text{for } x \in [-L/2, L/2]$$

$$= \cos(2\pi \frac{n x}{L}) + i \sin(2\pi \frac{n x}{L}), \quad i = \sqrt{-1}$$

We extend the domain periodically to $(-\infty, \infty)$

In this case, PDE solution must be extended to $(-\infty, \infty)$ or periodic boundary conditions are needed.

Ex: Inner product has the following properties.

$$i) \langle u, v \rangle = \langle v, u \rangle$$

$$ii) \langle \alpha u, v \rangle = \alpha \langle u, v \rangle = \langle u, \alpha v \rangle, \quad \alpha \text{ is scalar.}$$

$$iii) \langle u+v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$$

$$iv) \langle u, u \rangle = 0 \text{ if and only if } u = 0$$

$$\text{Ex: Show } \langle \sin 3x, \cos 3x \rangle = 0 \quad \text{for } x \in [-\pi, \pi]$$

Ex: Show or verify

$$\{1, \cos x, \cos 2x, \dots\} = \{\cos kx\}, k \geq 0$$

is an orthogonal set. for $x \in [-\pi, \pi]$

Ex: Show $\{e_1, e_2, e_3\}$ form a basis

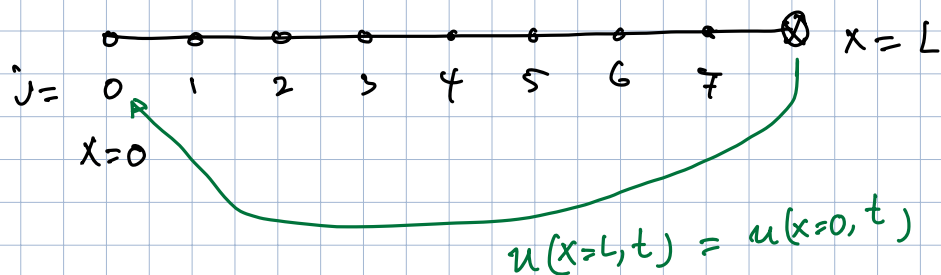
$$\text{where } e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Spectral discretization

Consider a grid of N points

$$\text{with } \Delta x = \frac{L}{N}, \quad x_j = j \Delta x = jL/N$$

$$j = 0, \dots, N-1$$



using $\phi_n(x) = e^{2\pi i n x / L}$

$$u_j(t) = u(x_j, t) = \sum_{n=-N/2+1}^{N/2} \hat{u}_n(t) e^{2\pi i n j L / N / L}$$

$\Rightarrow$ If we know $\hat{u}_n(t)$ we can compute $u_j(t)$
 $\nearrow$ Fourier coefficients

We say: $u_j(t) = F^{-1} \{ \hat{u}_n(t) \} = \sum_{n=-N/2+1}^{N/2} \hat{u}_n(t) e^{2\pi i n j / N}$

Also $\hat{u}_n(t) = F \{ u_j(t) \} = \frac{1}{N} \sum_{j=0}^{N-1} u_j(t) e^{-2\pi i j / N}$

be able to do:

Given $u_j(t)$ for $j = 0 \dots N-1$

compute $\hat{u}_n(t)$ by a package

Remark: numpy and scipy provides tools for FFT

H.W. (e.g. Lect #10)

$u^n = A u^{n-1}$ for a finite difference scheme

let A be a $N \times N$ matrix and $\{\phi_k\}$ are eigenvalues with corresponding eigen vector $\{\lambda_k\}$

Show $u^n = \sum_{k=1}^N C_k \lambda_k^n \phi_k$

Lecture #12, Oct 23, 2025

Read Appendix C (Eigenvalues and inner product)
(Leveque.)

Ex: $\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = 0$, a may be constant
or may be $a(x)$

Two options in spectral method.

A) find $\hat{u}_k(t)$ where $u(x_j, t) = \sum_{k=-N/2+1}^{N/2} \hat{u}_k(t) e^{2\pi i k j / N}$

(Galerkin method)

$$\begin{array}{cccccccc}
 j = 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\
 \circ & \circ & \circ & \circ & \circ & \circ & \circ & \circ \\
 k = -N/2 + 1 & & & & & & & x
 \end{array}$$

then take Inverse F.T. to get $u(x_j, t)$

B) Discretize $a \frac{\partial u}{\partial x}$ on $\{x_j\}$ points and

Solve for $u(x_j, t)$

Collocation
method-

Remark For non linear PDE, Galerkin method gives full matrix because product in physical space becomes convolution in Fourier space

$$\text{e.g. } u(x) * v(x) = \int_{-\infty}^{\infty} u(x) v(x-y) dy$$

Ex: i) show that

$$\left\langle e^{2\pi i n x / L}, e^{2\pi i m x / L} \right\rangle = \begin{cases} L & \text{if } n = m \\ 0 & \text{if } n \neq m \end{cases}$$

ii) Derive

$$\frac{\partial \hat{u}_k(t)}{\partial t} + a \left(\frac{2\pi i k}{L} \right) \hat{u}_k(t) = 0$$

$$\text{for } \frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = 0$$

$$\text{using } u(x, t) = \sum_{k=-N/2+1}^{N/2} \hat{u}_k(t) e^{2\pi i k x / L}$$

Numerical example:

$$\frac{u_j^{n+1} - u_j^{n-1}}{\Delta t} + a \frac{\partial u}{\partial x}(x_j, t_n) = 0$$

using Leap Frog method.

FFT normalization in numpy/matlab

$$\text{Expect } \|u\|_2^2 = \frac{1}{N} \|\hat{u}_k\|_2^2$$

Ex: Derive the stability condition

for solving $\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = 0$

using Leap-Frog scheme and Fourier spectral method.

Hint let $G = \frac{u^{n+1}}{u^n}$

show $G^2 + 2i\mu G - 1 = 0$, $\mu = ak\Delta t$

show $\Delta t \leq \frac{\Delta x}{C}$ to ensure $|G| < 1$

Find the value of C .

Here $C = \pi \max(a) \approx 3.14(1.2) \approx 3.5$

Ex: Convert the physical Cole to solve

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = 0$$

or $\frac{\partial u}{\partial t} + \frac{\partial}{\partial x} \left(\frac{1}{2} u^2 \right) = 0$

Ex: Compare Leap-Frog spectral solution
with Leap-Frog F.D solⁿ

Ex: 1) verify that Fourier modes e^{imx} and e^{inx} are orthogonal in $[0, 2\pi]$

Hint: $\int_0^{2\pi} e^{imx} e^{inx} dx = \begin{cases} 2\pi & m=n \\ 0 & m \neq n \end{cases}$

2) Why the set $\{e^{imx}\}_{m=-\infty}^{\infty}$

is mutually orthogonal but not orthonormal.

3) $u(x) = \sum_{k=-\infty}^{\infty} c_k e^{ikx}$

Show that $c_k = \frac{1}{2\pi} \langle u, e^{ikx} \rangle$

4) Consider 1D wave equation with $a = -1$

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = 0$$

let $U = [u_0 \dots u_{2N-1}]^T$ be the solution vector on $\{x_j\}$ grid for $j = 0, \dots, 2N-1$

let D be a matrix such that

DU is the vector containing $\frac{\partial u}{\partial x}$

on $\{x_j\}$.

Show that D is skew symmetric

Hint

$$\frac{dU}{dt} = DU$$

$$\frac{d}{dt} U^T U = 2 U^T D U$$

By conservation, $\frac{d}{dt} U^T U = 0$

$$\text{so } 2 U^T D U = U^T D U = -U^T D U$$

$$U^T D U = (U^T D U)^T = (D U)^T U$$

$$= U^T D^T U$$

$$= -U^T D U$$

Thus $\boxed{D^T = -D}$